DHCP Snooping – Cisco Packet Tracer Lab

1. Objective

The purpose of this lab is to configure DHCP Snooping on a Cisco Layer 2 switch to protect a network against unauthorized (Rogue) DHCP servers.

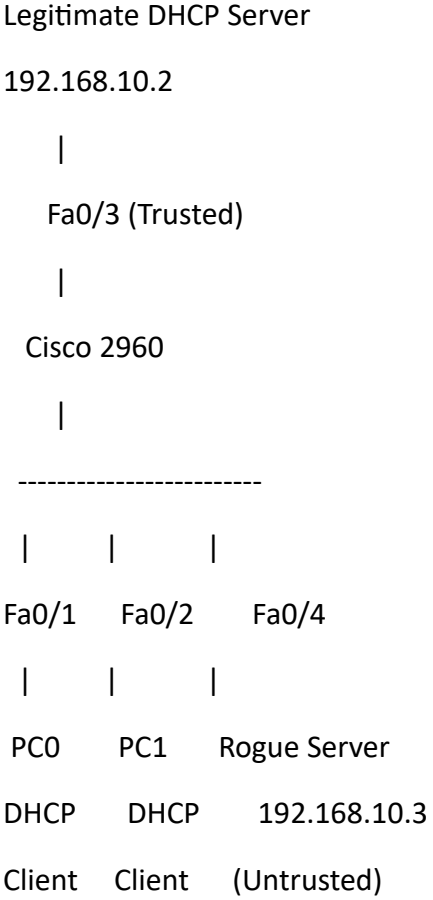
DHCP Snooping allows DHCP server messages from trusted ports and blocks unauthorized DHCP server messages received on untrusted ports.

2. Lab Topology

Devices Used:

- 1 × Cisco 2960 Switch
- 2 × Server-PT
- 2 × PC-PT
- 4 × Copper Straight-Through Cables

Network Diagram:



3. IP Addressing

Device	IP Address	Role
Legitimate Server	192.168.10.2	Authorized DHCP
Rogue Server	192.168.10.3	Unauthorized DHCP
PC0	DHCP	Client
PC1	DHCP	Client

Subnet Mask: 255.255.255.0

4. Configure the Legitimate DHCP Server

1. Open the Top Server.
2. Navigate to **Desktop → IP Configuration**.
3. Configure:

IP Address: 192.168.10.2

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

4. Navigate to **Services → DHCP**.
5. Turn DHCP Service ON.
6. Create the following pool:

Setting	Value
Pool Name	LEGIT_DHCP
Default Gateway	0.0.0.0
DNS Server	8.8.8.8
Start IP	192.168.10.100
Subnet Mask	255.255.255.0
Maximum Users	50

7. Click **Add**.

Note: Packet Tracer may contain a default serverPool. In this lab, its Remove button was disabled. We changed its address range to 192.168.20.10 to avoid overlap with the legitimate pool.

5. Configure the Rogue DHCP Server

1. Open the Right Server.
2. Navigate to **Desktop → IP Configuration**.
3. Configure:

IP Address: 192.168.10.3

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

4. Navigate to **Services → DHCP**.
5. Turn DHCP Service ON.
6. Create this pool:

Setting	**Value**
Pool Name	ROGUE_DHCP
Default Gateway	0.0.0.0
DNS Server	1.1.1.1
Start IP	192.168.10.200
Subnet Mask	255.255.255.0
Maximum Users	50

7. Click **Add**.

This server represents an unauthorized DHCP server.

6. Configure DHCP Snooping on the Cisco 2960

Open the switch CLI and enter:

enable

configure terminal

ip dhcp snooping

ip dhcp snooping vlan 1

end

Explanation:

- ip dhcp snooping – Enables DHCP Snooping globally.
- ip dhcp snooping vlan 1 – Enables DHCP Snooping on VLAN 1.

7. Configure the Trusted Port

The legitimate DHCP server is connected to FastEthernet0/3.

Configure this port as trusted:

enable

configure terminal

interface fastethernet 0/3

ip dhcp snooping trust

end

Important: Only the legitimate DHCP server port is trusted.

All other ports remain untrusted by default, including the rogue server port.

8. Verify DHCP Snooping Configuration

Run:

show ip dhcp snooping

Expected result:

Switch DHCP snooping is enabled

DHCP snooping is configured on following VLANs:

1

Interface Trusted

FastEthernet0/3 yes

This confirms that DHCP Snooping is enabled and Fa0/3 is trusted.

9. Configure DHCP Clients

For PC0 and PC1:

1. Open Desktop.
2. Select IP Configuration.
3. Click DHCP.

Expected addresses:

Device	Assigned IP
PC0	192.168.10.100
PC1	192.168.10.101

10. Verify Client IP Configuration

On each PC, open Command Prompt:

ipconfig /all

Observed results:

Details	PC0	PC1
IPv4 Address	192.168.10.100	192.168.10.101
Subnet Mask	255.255.255.0	255.255.255.0
DHCP Server	192.168.10.2	192.168.10.2
DNS Server	8.8.8.8	8.8.8.8

11. Verify DHCP Snooping Binding Table

On the switch:

show ip dhcp snooping binding

The switch learned the following client bindings:

MAC Address	IP Address	Interface
00:40:0B:75:17:13	192.168.10.100	Fa0/1
00:01:63:6D:33:0B	192.168.10.101	Fa0/2

The binding table records client MAC addresses, assigned IP addresses, VLANs, and switch interfaces.

12. Test DHCP Lease Renewal

On PC0:

ipconfig /release

ipconfig /renew

ipconfig /all

Observed result after renewal:

IPv4 Address: 192.168.10.102

DHCP Server: 192.168.10.2

DNS Server: 8.8.8.8

The client received another address from the legitimate DHCP server.

13. Important Verification Commands

show ip dhcp snooping

show ip dhcp snooping binding

show ip dhcp snooping statistics

show running-config

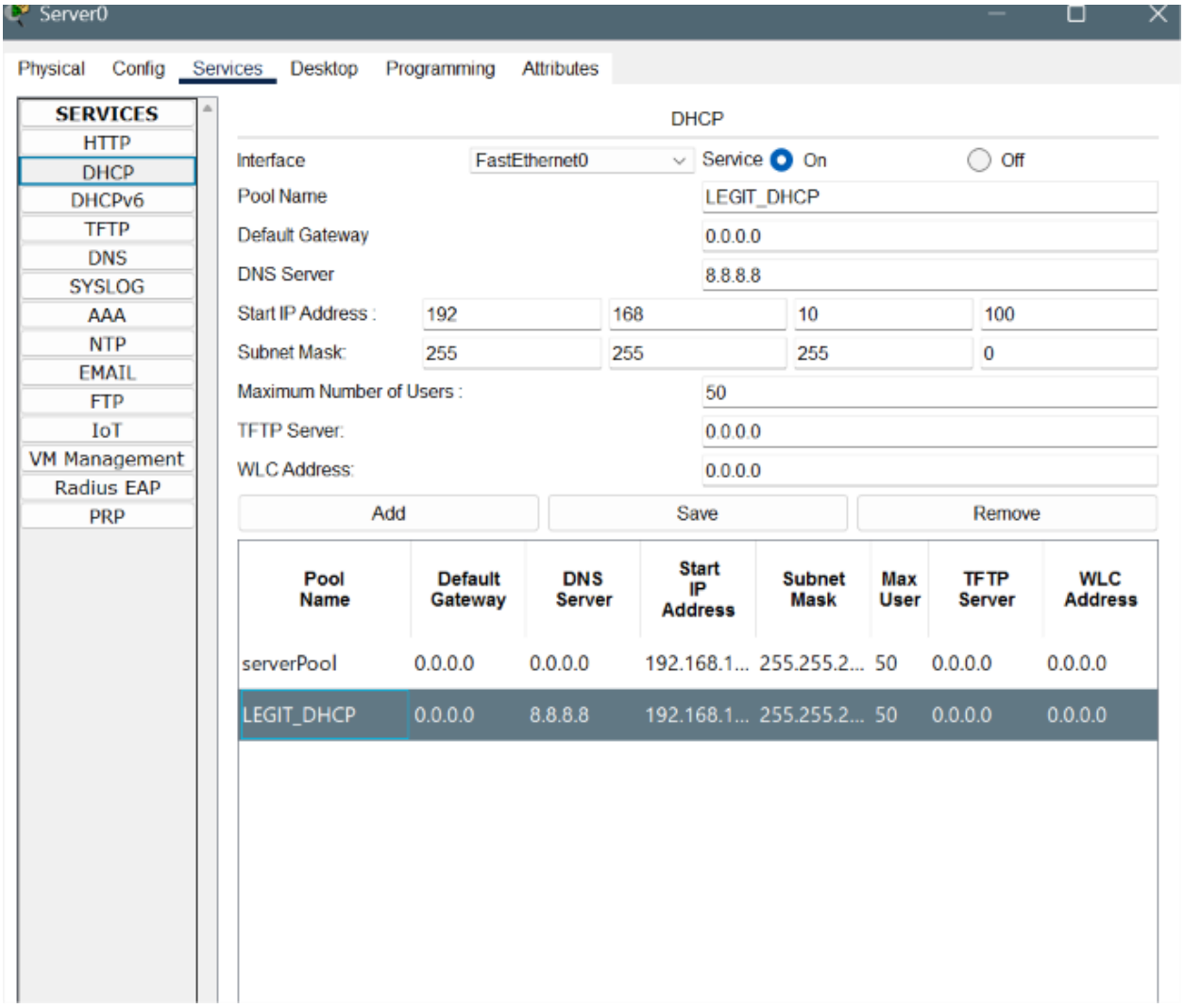
14. Expected Security Behavior

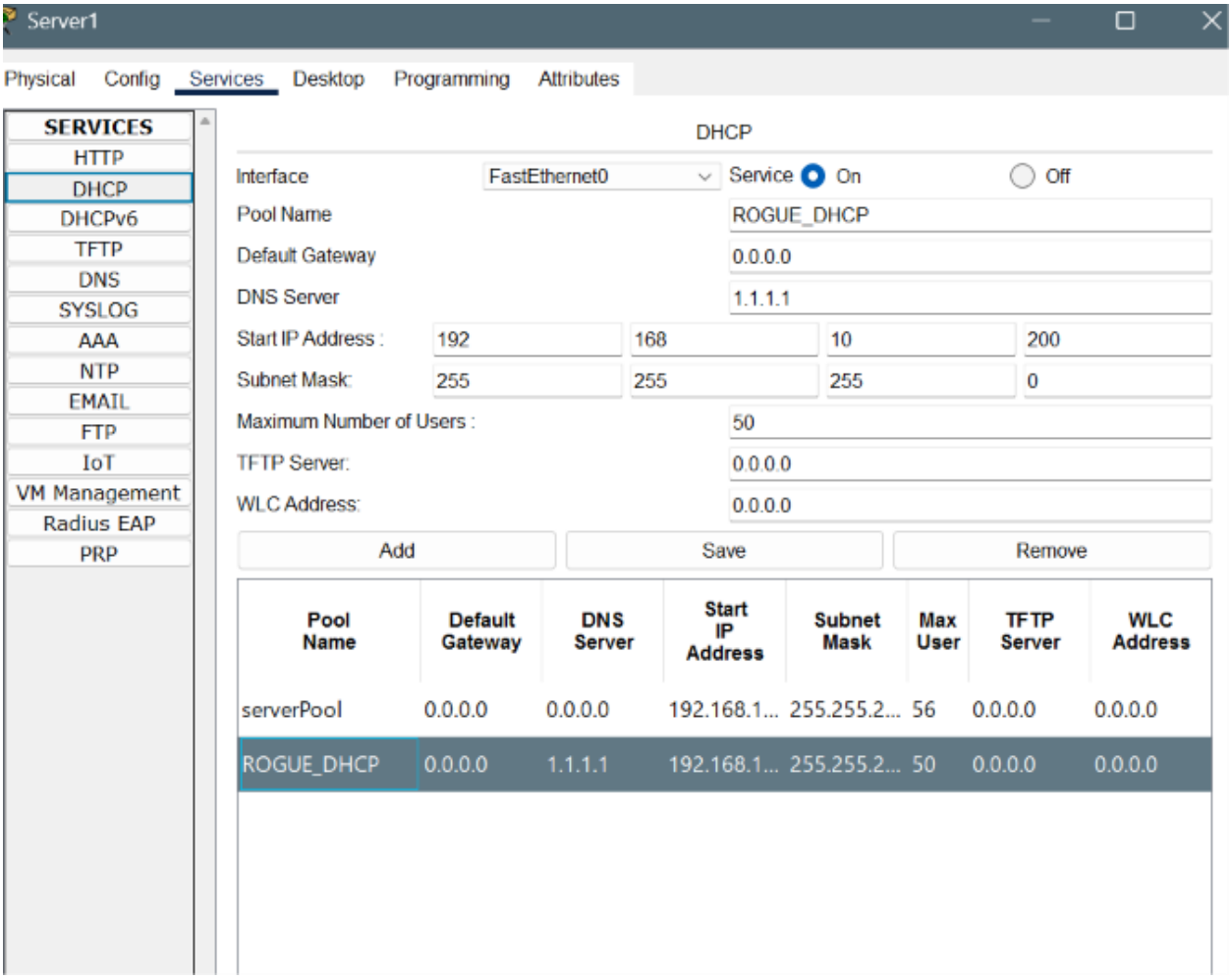
Port	Connected Device	Trust Status
Fa0/1	PC0	Untrusted
Fa0/2	PC1	Untrusted
Fa0/3	Legitimate DHCP Server	Trusted

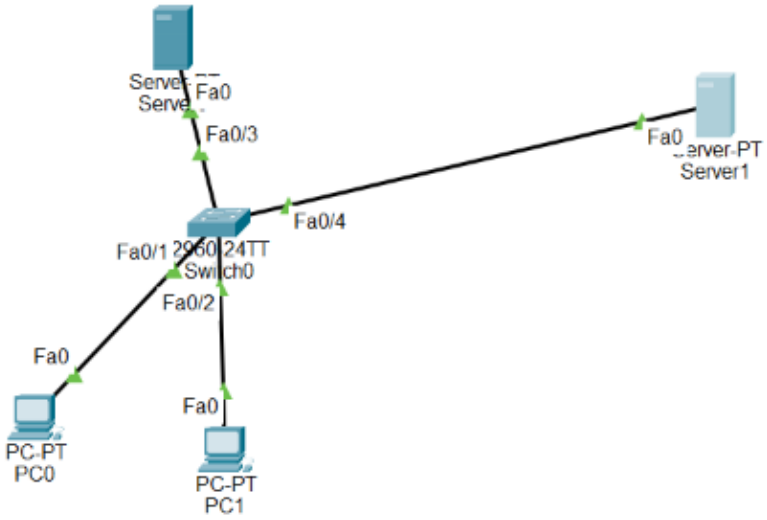
Fa0/4 Rogue DHCP Server Untrusted

DHCP server messages such as DHCPOFFER and DHCPACK received on untrusted ports are dropped by DHCP Snooping.

Note: The lab configuration and legitimate DHCP operation were verified. A separate rogue-server packet-drop test was not conclusively demonstrated in the recorded results. This should be verified in Simulation Mode before claiming the attack was successfully blocked.


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
LEGIT_DHCP
Default Gateway
0.0.0.0
DNS Server
8.8.8.8
Start IP Address :
192
168
10
100
Subnet Mask:
255
255
255
0
Maximum Number of Users :
50
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
0.0.0.0
0.0.0.0
192.168.1...
255.255.2...
50
0.0.0.0
0.0.0.0
LEGIT_DHCP
0.0.0.0
8.8.8.8
192.168.1...
255.255.2...
50
0.0.0.0
0.0.0.0


Server1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
ROGUE_DHCP
Default Gateway
0.0.0.0
DNS Server
1.1.1.1
Start IP Address :
192
168
10
200
Subnet Mask:
255
255
255
0
Maximum Number of Users :
50
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
0.0.0.0
0.0.0.0
192.168.1...
255.255.2...
56
0.0.0.0
0.0.0.0
ROGUE_DHCP
0.0.0.0
1.1.1.1
192.168.1...
255.255.2...
50
0.0.0.0
0.0.0.0


Fa0
Fa0/3
Fa0
Server1
Fa0/4
Fa0/1
Switch0
Fa0/2
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1

15. What I Learned

- Understanding DHCP Snooping.
- Difference between trusted and untrusted ports.
- Configuring DHCP Snooping on a Cisco 2960 switch.
- Configuring legitimate and rogue DHCP servers.
- Understanding DHCP client IP assignment.
- Verifying DHCP bindings.
- Troubleshooting incorrect DHCP pool assignments.
- Understanding how Layer 2 switches help protect networks against rogue DHCP servers.

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16. Interview Questions and Answers

Q1. What is DHCP Snooping?

DHCP Snooping is a Layer 2 security feature that protects a network against unauthorized DHCP servers.

Q2. What is a Rogue DHCP Server?

A Rogue DHCP Server is an unauthorized DHCP server that provides incorrect IP configuration to network clients.

Q3. What is a Trusted Port?

A trusted port is a switch interface allowed to receive DHCP server messages.

Q4. What is an Untrusted Port?

An untrusted port is an interface where DHCP server messages are blocked. Client DHCP requests are allowed.

Q5. Which port did you configure as trusted in your lab?

FastEthernet0/3, because the legitimate DHCP server was connected to that port.

Q6. Why did you keep Fa0/4 untrusted?

Because the rogue DHCP server was connected to Fa0/4. We wanted to prevent unauthorized DHCP offers from entering the network.

Q7. What is a DHCP Snooping Binding Table?

It stores DHCP client information, including MAC address, assigned IP address, VLAN, and switch interface.

Q8. Which command verifies DHCP Snooping?

show ip dhcp snooping

Q9. Which command displays DHCP bindings?

show ip dhcp snooping binding

Q10. What is DHCP DORA?

DORA is the four-step DHCP process:

1. Discover
2. Offer
3. Request
4. Acknowledgment

Q11. What happens when a Rogue DHCP Server sends a DHCP Offer through an untrusted port?

DHCP Snooping drops the unauthorized server message.

Q12. Is DHCP Snooping a Layer 2 or Layer 3 security feature?

It is a Layer 2 security feature implemented on switches.

Q13. Does DHCP Snooping block DHCP client requests?

No. DHCP client requests are allowed on untrusted ports. The feature filters unauthorized DHCP server messages.

Q14. Why is DHCP Snooping important in enterprise networks?

It prevents unauthorized DHCP servers from distributing incorrect IP addresses, gateways, and DNS settings to clients.

Q15. Which Cisco command enables DHCP Snooping on VLAN 1?

ip dhcp snooping vlan 1

Conclusion

In this lab, I configured DHCP Snooping on a Cisco 2960 switch, created legitimate and rogue DHCP servers, configured trusted and untrusted ports, assigned IP addresses to clients, and verified DHCP Snooping bindings.

This practical exercise helped me understand how Layer 2 security protects enterprise networks from unauthorized DHCP servers